BAIRD

Technology

Do You Believe in Magic (Numbers to Measure Sales Efficiency)?

In light of the current environment, we thought it would be beneficial to refresh the Magic Numbers for SaaS names. We have added more granularity to adjust for cycle lengths. We recognize that our initial definition of the Magic Number, a metric used to assess sales efficiency in SaaS models, was based on quarterly sequential growth and may not adequately account for longer sales cycles. As a result, we've included a revised magic number here, along with the historical data, year-over-year comparison for C4Q21, percentile breakdown, and trends by coverage.

- **What is Magic Number?** It's a measure of sales efficiency in a SaaS model. The calculation is an approximation of the revenue generated (TTM average) for each dollar spent on S&M in the preceding cycle. Though there is no hard rule on interpreting Magic Numbers, general consensus is a Magic Number greater than 0.75-1.0 indicates that a company can make additional S&M capital investments; lower indicates then company should prioritize S&M efficiency improvement.
- **Formula notes.** We use subscription revenue for companies that report it separately; otherwise, we use total revenue. The general formula used to calculate the MN is as follows: ([Subscription] Revenue Growth %)/(Prior Cycle S&M spend as a % of [Subscription] Revenue). Expanding the formula fully yields:
  - Shorter Sales-Cycles (sub-90 Days): (Quarterly [Subscription] Revenue - Previous Quarter's [Subscription] Revenue) * 4 /(Previous Quarter's S&M Spend). We then take trailing four-quarter average.
  - Longer Sales-Cycles: (TTM [Subscription] Revenue - Prior Year's [Subscription] Revenue) / (Prior Year's S&M Spend).
- **Sector-specific commentary.**
  - **Oliver (Software & SaaS).** Product-led growth (PLG) continues to be a dominant theme in our group, representing a shift away from traditional, heavy sales-driven models. TEAM (top large-cap pick) maintains the highest magic number on our list and has demonstrated further substantial efficiency gains. SPT/AMPL/FRSH are the other PLG emerging leaders; all of them, with the exception of SPT (which has begun to deliver on leverage), remain in a high-investment mode at the moment, as they are in the early stages of their growth.
  - **Power (Cloud Software).** ZM, DOCU and FIVN's magic numbers have all declined from Q2 levels, driven by a combination of tough comps and moderating revenue growth. RNG and EVBG's magic numbers have remained relatively consistent over the past few quarters, though still below the leaders.
  - **Ruykhaver (Security & Infrastructure Software).** DDOG currently has the highest magic number in our coverage, as the company has continued to see the business environment improve and has now lapped weaker results. CRWD, ZS, and OKTA all hold magic numbers above the broader cohort median, while NET remains in high-investment mode.
  - **Vruwink (Vertical Software).** Vertical software companies serve as strategic partners to a dedicated industry. Efficiency in sales and marketing is achieved by targeting a specific audience, focusing on customer success to earn and build loyalty, and then utilizing this experience to drive a reference selling model for the future. In our coverage, VEEV ranks highest in this metric.

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Please refer to Appendix - Important Disclosures and Analyst Certification

Details

Historical Data table

The table below gives historic Magic Number data back to C1Q20. Results are sorted by Magic Number in C4Q21.

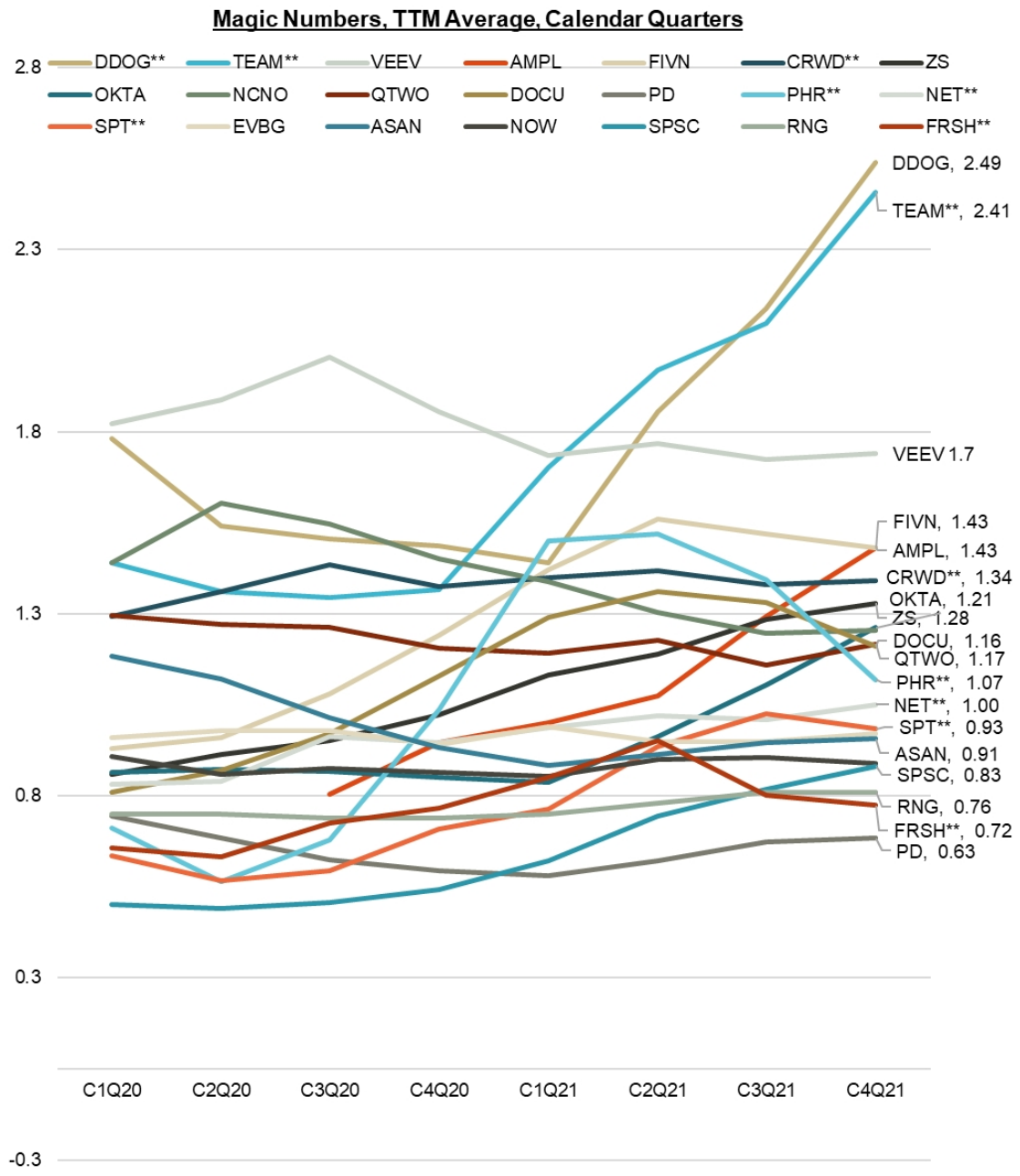
Magic Numbers, TTM Average, Calendar Quarters								
Company	C1Q20	C2Q20	C3Q20	C4Q20	C1Q21	C2Q21	C3Q21	C4Q21
ZM	2.71	7.00	11.62	16.29	17.35	13.20	8.55	3.67
DDOG**	1.73	1.49	1.46	1.44	1.39	1.80	2.09	2.49
TEAM**	1.39	1.31	1.30	1.32	1.65	1.92	2.05	2.41
VEEV	1.77	1.84	1.95	1.81	1.68	1.72	1.67	1.69
AMPL			0.76	0.90	0.95	1.02	1.24	1.43
FIVN	0.88	0.91	1.03	1.19	1.37	1.51	1.47	1.43
CRWD**	1.24	1.31	1.38	1.32	1.35	1.37	1.33	1.34
ZS	0.81	0.86	0.90	0.97	1.08	1.14	1.24	1.28
OKTA	0.82	0.82	0.82	0.80	0.79	0.91	1.06	1.21
NCNO	1.39	1.55	1.50	1.40	1.34	1.25	1.20	1.20
QTWO	1.25	1.22	1.21	1.16	1.14	1.18	1.11	1.17
DOCU	0.76	0.82	0.92	1.08	1.24	1.31	1.28	1.16
PHR**	0.66	0.52	0.63	0.99	1.45	1.47	1.34	1.07
NET**	0.78	0.79	0.91	0.90	0.94	0.97	0.96	1.00
SPT**	0.58	0.52	0.54	0.66	0.71	0.88	0.98	0.93
EVBG	0.91	0.93	0.93	0.89	0.94	0.90	0.90	0.92
ASAN	1.13	1.07	0.97	0.88	0.83	0.86	0.90	0.91
NOW	0.86	0.81	0.82	0.81	0.80	0.85	0.86	0.84
SPSC	0.45	0.44	0.46	0.49	0.57	0.69	0.77	0.83
RNG	0.70	0.70	0.69	0.69	0.70	0.73	0.76	0.76
FRSH**	0.61	0.58	0.68	0.72	0.80	0.90	0.75	0.72
PD	0.69	0.64	0.57	0.55	0.53	0.57	0.62	0.63
WK	0.55	0.49	0.47	0.45	0.45	0.51	0.58	0.62
Median	0.84	0.84	0.91	0.90	0.95	1.02	1.11	1.16

Analyst:
Oliver
Power
Ruykhaver
Vruwink

***Note: SaaS cos with short sales-cycle model (sub-90 days) are based on the original quarter-over-quarter magic number. For off-calendar quarters, closest quarter is used; e.g., for C4Q21, using quarter ended January, 2022.*

Source: Company Documents, Baird Research

Historical Chart (Minus ZM for Readability)



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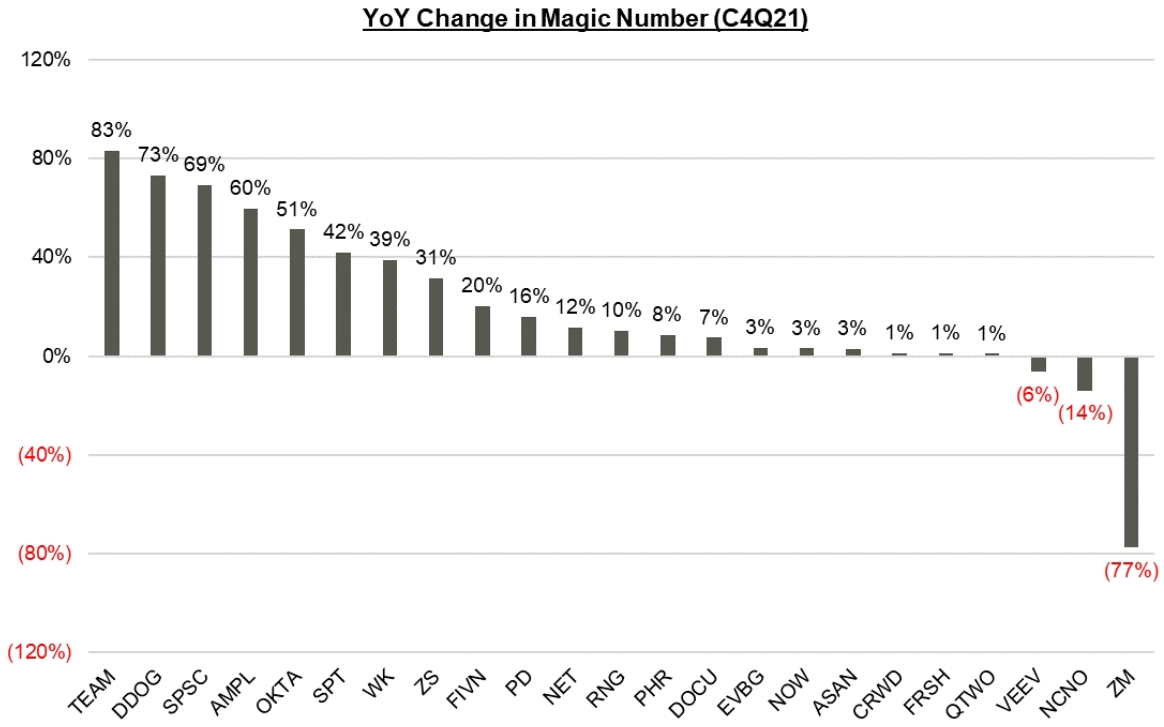
Source: Company Documents, Baird Research

C4Q21 YoY Change in Four-Quarter Average Magic Number
The graph below displays the YoY change in our Magic Number Calculation from C1Q20 to C4Q21.

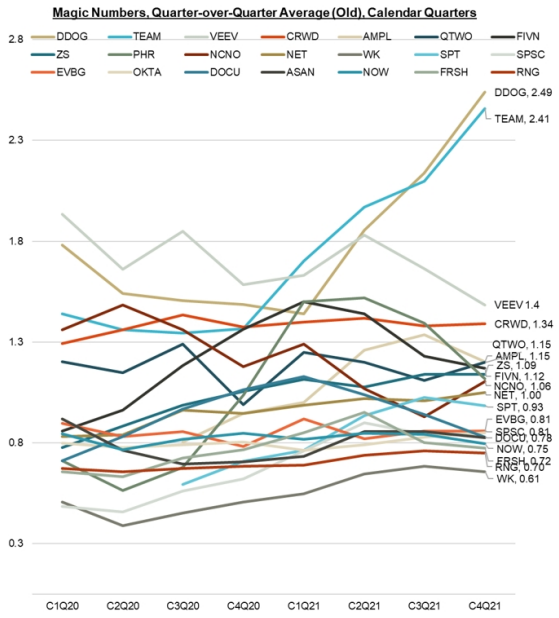


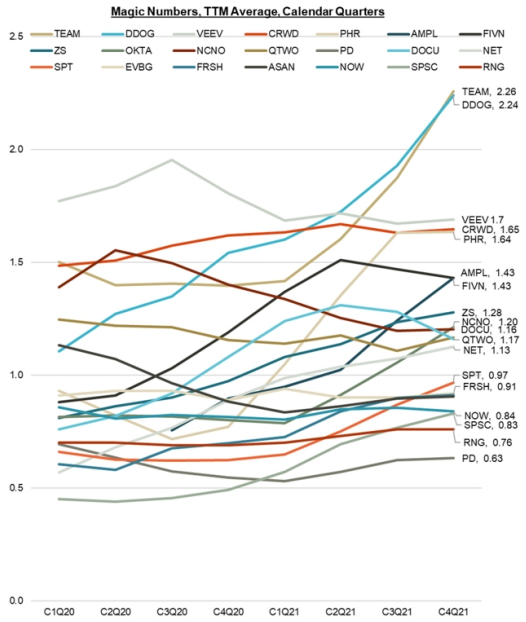
***Note: SaaS cos with short sales-cycle model (sub-90 days) are based on the original quarter-over-quarter magic number. For off-calendar quarters, closest quarter is used; e.g., for C4Q21, using quarter ended January, 2022.*
Source: Company Documents, Baird Research

Comparison of Year-Over-Year Annualized Metric with Quarter-Over-Quarter Sequential Metric:
The following charts compare the sequential qr-qr magic number (legacy definition) to the TTM annualized magic number (revised definition) across all software sub-sectors.

For names with relatively longer sales and deployment cycles, the revised version of magic numbers is more representative of the actual ROI generated by sales motions due to the lag effect between investments and the resulting incremental top-line growth — the qr-qr sequential metric captures trends that are not indicative of reality in these cases. The new definition smoothes out quarterly perturbations that are merely noise around the trendlines in the shown time horizons, while not missing any actual inflections and still capturing cumulative quarterly trends over the same time horizon.

Our original q/q sequential metric is more appropriate for shorter cycle/PLG names with cycle lengths under 90 days (intra-quarter)

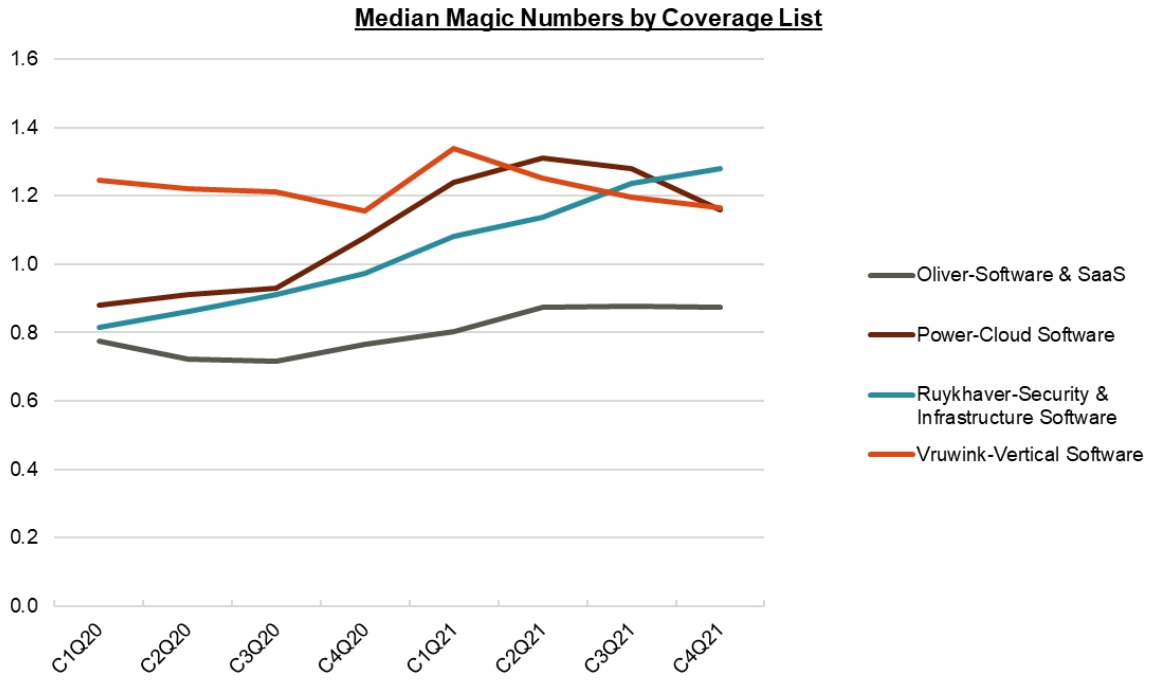




Note: For off-calendar quarters, closest quarter is used; e.g., for C4Q20, using quarter ended January, 2021.
Source: Company Documents, Baird Research

Median Magic Numbers by Coverage List

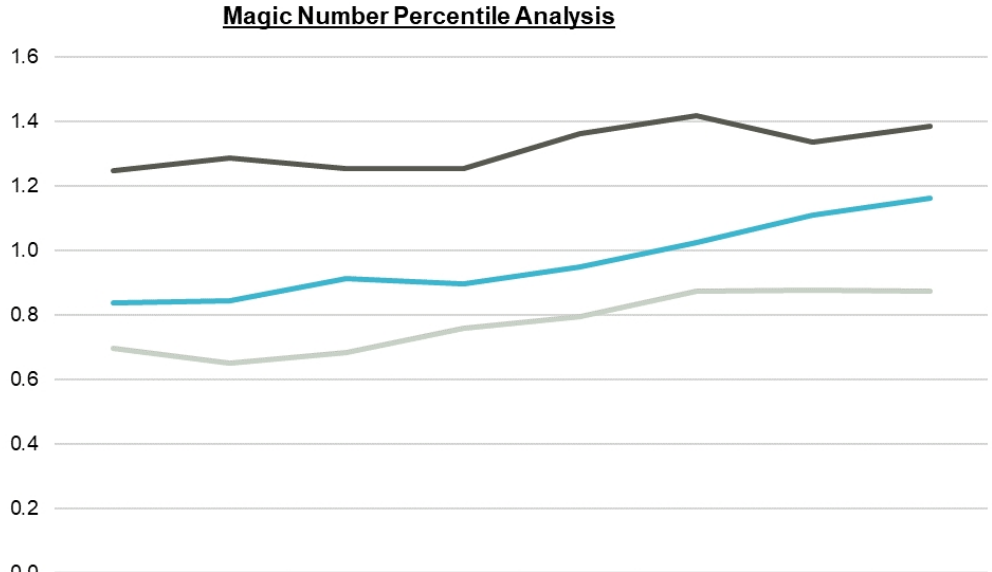
The graph below shows median magic numbers of the companies included in the above analysis split out for each analyst's coverage.



Note: For off-calendar quarters, closest quarter is used; e.g., for C4Q20, using quarter ended January, 2021.
Source: Company Documents, Baird Research

Magic Number Percentile Analysis

The graph below shows the progression of the 75th, 50th, and 25th percentiles of the cohort.



	C1Q20	C2Q20	C3Q20	C4Q20	C1Q21	C2Q21	C3Q21	C4Q21
75th Percentile	1.24	1.29	1.25	1.25	1.36	1.42	1.34	1.39
50th Percentile	0.84	0.84	0.91	0.90	0.95	1.02	1.11	1.16
25th Percentile	0.70	0.65	0.68	0.76	0.79	0.87	0.88	0.87

Note: For off-calendar quarters, closest quarter is used; e.g., for C4Q21, using quarter ended January, 2022.
Source: Company Documents, Baird Research

Valuation & Risks

Zoom Video Communications, Inc. (ZM)

Valuation

Our $140 target price is based on 30x our F2023 (C22) FCF estimate, below the larger FCF SaaS names at about 40x, reflecting moderating revenue growth despite higher margins.

Risks

Risks include rising competition, new products and services, video concentration, COVID-19 puts and takes, and valuation.

Datadog, Inc. (DDOG)

Valuation

We derive our $190 price target using a 2023E EV/Sales multiple of 31.6x, a premium to a peer group trading at ~16.5x CY23E EV/Sales. We believe this multiple is warranted given Datadog's attractive positioning in a rapidly growing, though competitive, market, platform opportunity, and a substantial growth opportunity.

Risks

Risks include, but are not limited to, competition, macroeconomic pressures on spend, and failure to gain adoption of new products.

Veeva Systems Inc. (VEEV)

Valuation

Our $288 target price is supported by a multi-year DCF as well as 18x P/Sales and 45x P/FCF on NTM+1 estimates, at the high end of peer multiples (8-19x NTM P/Sales, 25-60x NTM P/FCF) given above-peer expectations for revenue growth and profitability.

Risks

Key risks: variability in quarterly financial results/guidance, the ability to win new customers and expand product adoption, life sciences market fundamentals and the impact on customer spending decisions, competition, and customer concentration.

Five9, Inc. (FIVN)

Valuation

Our $125 target price is based on 12x our 2022 revenue forecast, in line with the SaaS cohort at ~12x, reflecting strong growth offset by lower gross margins.

Risks

Risks include competition, M&A integration, COVID impact uncertainties, high investor expectations and valuation at the higher end of historical levels.

CrowdStrike Holdings, Inc. (CRWD)

Valuation

Our $275 price target implies a CY23E EV/Sales multiple of ~23x. This multiple is slightly higher than the median for high-growth SaaS peers (~19x EV/Sales), which we believe is justified by strong positioning relative to legacy peers, successful platform adoption and strong execution at scale.

Risks

Risks include strong competition in the endpoint protection market, risks from entering into adjacent markets, and pricing/asp risks, among others.

Atlassian Corporation Plc (TEAM)

Valuation

Target of $520 represents ~60x FCF at the end-of-transition (FY25E), implying ~27x revenues FY24E, at the high end of quality high-growth SaaS peers trading in the 15x-30x range. We feel multiple is justified given uniquely efficient go to-market, strong management, and rare combination of size, growth, and profitability.

Risks

Risks include: lack of visibility from online distribution and marketing model, premium valuation, ability to expand outside of core software developer user group, and competition.

Q2 Holdings, Inc. (QTWO)

Valuation

Our $92 target price is supported by a multi-year DCF as well as 7x P/Sales on NTM+1 estimates, within the range of peer multiples (7-16x NTM P/Sales). We believe this is reasonable in the context of the company's forward revenue and EBITDA growth opportunity.

Risks

Key risks: variability in quarterly financial results/guidance, the ability to win new business, market fundamentals for RCFI customers, competition, the uncertain environment created by COVID-19 and impacts on quarterly financial performance.

Zscaler, Inc. (ZS)

Valuation

Our $300 price target is supported by a DCF analysis and implies a CY23E EV/Sales multiple of ~25.6x. This multiple is a premium to the median for high-growth SaaS peers (~17x EV/Sales), which we believe is justified by the company's strong offering and positioning in a growing market.

Risks

Risks include high competition in the network security industry, especially among cloud offerings; a history of losses/ minimal non-GAAP profitability, and macroeconomic risks.

nCino, Inc. (NCNO)

Valuation

Our $55 target price is supported by a multi-year DCF as well as 11x P/Sales on NTM+1 estimates, at the high end peer multiples (6-14x NTM P/Sales) given the company’s premium revenue growth opportunity.

Risks

Key risks: industry health, timeline around investment decisions, growing new revenue streams, future stock sales, high expectations.

DocuSign, Inc. (DOCU)

Valuation

Our $95 target price is based on 8x our F23 revenue forecast (C22), slightly below the SaaS group at ~9x C22, reflecting a strong competitive position and profitability offset by decelerating growth due to tough comps.

Risks

Risks include dominated by one product, international expansion, COVID-19 puts and takes, competition and valuation.

Cloudflare, Inc. (NET)

Valuation

Our $130 price target represents a 2023E EV/Sales multiple of 34.0x, vs. peers trading at a median ~19.5x. We believe Cloudflare is in the early stages of capturing what we see as a large market opportunity, justifying a premium to peers.

Risks

Risks include competition in both the performance and security markets, failure to gain traction with new products, reputational risk from its client base, and a potential deterioration of the JD partnership.

Sprout Social, Inc. (SPT)

Valuation

Our $110 price target reflects EV/Sales of ~18x our revised FY23 revenue estimates, compares to the peer group average multiple of ~12x. We believe the premium multiples is justified due to large addressable market, early stages of opportunity, founder-led vision, ability to scale margins and FCF over time.

Risks

Risks include: Contract/Platform risk (social moves quickly), SMB exposure/higher churn, competition, need to ramp profitability over time.

ServiceNow, Inc. (NOW)

Valuation

Our $600 price target represents EV/FCF multiple of 53x FY22, on our revised estimates, comparing to FCF growth rate of 30%+ and three-year average multiple of 41x. We believe premium multiple is deserved, given: 1) above-average growth-profile; 2) strong competitive-positioning; 3) large TAM; and 4) top-decile FCF margin.

Risks

Risks include: valuation as NOW is one of the most expensive stocks in the SaaS universe, competition, and large deal closures slowing in uncertain macro environment.

SPS Commerce, Inc. (SPSC)

Valuation

Our $156 target price is supported by a multi-year DCF as well as 11x P/Sales and 47x P/FCF on NTM+1 estimates, within the range of peer multiples (7-14x NTM P/Sales, 25-60x NTM P/FCF), warranted considering revised growth potential and the opportunity to improve FCF margin over coming years.

Risks

Key risks: variability in quarterly financial performance, cyclicality in retail industry, the ability to recruit new customers, retaining sales personnel, competition, the uncertain environment created by COVID-19 and impacts on quarterly financial performance.

Asana, Inc. (ASAN)

Valuation

Our $65 price target, implying an EV/Sales multiple of ~20x our revised CY22 revenue estimates is at a premium to comparable high-growth peer group average multiples of 14x. We believe the multiple is justified due to large unpenetrated TAM, scalable margins and FCF.

Risks

Risks include incumbents, new competition from disruptors and valuation.

Everbridge, Inc. (EVBG)

Valuation

Our $45 target price is based on 5x our 2022 revenue forecast. That puts it below the broader UCC group at ~6x, reflecting slowing growth and the recent leadership transition.

Risks

Risks include competitive threats, management changes, pricing, limited profitability, international expansion and COVID-19 uncertainties.

Okta, Inc. (OKTA)

Valuation

Our $195 price target is supported by a DCF analysis and implies a CY23E EV/Sales multiple of ~14.1x. Okta's multiple is discounted vs. the median for high-growth SaaS (~20.5x EV/Sales), which we believe is justified given strong positioning but organic growth which had slowed and Auth0 execution risk.

Risks

Risks include increasing competition in the prized IAM market, including from vendors not traditionally focused on the space such as MSFT and AMZN, and a heightened valuation which could compress sharply should growth disappoint.

RingCentral, Inc. (RNG)

Valuation

Our $130 target price is based on 7x our 2022 revenue forecast, below the SaaS group at ~9x, driven by continued concerns around competition offset somewhat by accelerating growth.

Risks

Risks include competitive threats, international expansion, managing partner relations and limited profitability.

Workiva, Inc. (WK)

Valuation

Our $150 price target represents 14x FY22 Revenue estimates, and compares to a broader group of SaaS comps with average P/Sales multiple of ~10x; we view premium as justified based on large installed base in SEC and growth in non-SEC use-cases such as statutory reporting, GRC, ESG.

Risks

Risks include: inability to expand beyond core SEC use cases, negative margins, and new competition entering the market.

PagerDuty Inc. (PD)

Valuation

Our $47 price target, implying an EV/Sales multiple of ~10x our FY23 estimates, is a premium to comparable growth-profile peer group median multiple at ~8.5x. We believe the combination of better sales execution, spending environment and a large green field opportunity for PagerDuty could contribute to multiple expansion.

Risks

Risks include: competition from vendors including OpsGenie (Atlassian) and VictorOps (Splunk); developing market that PagerDuty is pioneering (i.e., less established); and lack of profitability.

Salesforce.com, Inc. (CRM)

Valuation

Our $300 price target implies EV/Sales multiple of ~9x CY23E, compares to the median P/Sales multiple of 7x commanded by a comparable group of growth Software stocks. We believe this valuation is warranted given CRM's dominant size, sustained >20% organic revenue growth, industry leadership, and strong management execution.

Risks

Salesforce.com has consistently grown >20%+ and has now reached >$21B in revenue. Growth becomes difficult at that pace and scale; business could slow (particularly given macro concerns). Salesforce.com also competes with large, relevant players.

BlackLine, Inc. (BL)

Valuation

Our $105 price target reflects paring back of some near-term expectations while we wait for growth reacceleration, and represents an EV/Sales multiple of 12x 2022E sales which is close to consensus number. We believe the premium valuation is justified by BlackLine's large and unpenetrated TAM and first-mover advantage.

Risks

Risks include sales execution and customer success hiccups, premium valuation, longer sales cycle on strategic products (Intercompany Hub, Smart Close), and potential competition from variety of players.

Amplitude, Inc. (AMPL)

Valuation

Our $40 price target implies an EV/Sales multiple of ~17x our revised CY22 revenue estimate and compares to the peer-group median multiple of ~14x. We believe the premium multiple is justified due to the large addressable-market, early stages of digital optimization with little direct competition and strong management.

Risks

Risks include the trajectory of the new category, moving from Periphery to Core, Competition, and Valuation.

Phreesia, Inc. (PHR)

Valuation

Our $51 target price is supported by a multi-year DCF as well as 7x P/Sales on NTM+1 estimates, within the range of triangulated peer multiples (Payments: 4-7x NTM P/Sales, Vertical Software: 7-16x NTM P/Sales), balancing the company's revenue growth opportunity with potential execution risk on recently accelerated investments.

Risks

Key risks: valuation multiples, competition, and information security.

Freshworks, Inc. (FRSH)

Valuation

Our $40 price target represents an EV/Sales multiple of ~20x our CY22 revenue estimates compared to the peer-group median multiples of ~15x. We believe the premium multiple is justified due to the large addressable market, early stages of that opportunity, accelerating mid-market, high-velocity land-and-expand motion, strong founder-led management

Risks

Risks include competition, profitability, geographic/employee concentration, execution, and valuation.

Appendix - Important Disclosures and Analyst Certification

Approved on 26 April 2022 00:27EDT/ Published on 26 April 2022 01:05EDT.

Covered Companies Mentioned

All stock prices below are the 4/25/2022 closing price.

Amplitude, Inc. (AMPL - $18.68 - Neutral)
Asana, Inc. (ASAN - $30.54 - Outperform)
Atlassian Corporation Plc (TEAM - $253.63 - Outperform)
BlackLine, Inc. (BL - $68.16 - Outperform)
Cloudflare, Inc. (NET - $98.71 - Neutral)
CrowdStrike Holdings, Inc. (CRWD - $214.58 - Outperform)
Datadog, Inc. (DDOG - $121.68 - Neutral)
DocuSign, Inc. (DOCU - $89.95 - Neutral)
Everbridge, Inc. (EVBG - $47.18 - Neutral)
Five9, Inc. (FIVN - $106.25 - Outperform)
Freshworks, Inc. (FRSH - $18.82 - Outperform)
Okta, Inc. (OKTA - $138.40 - Neutral)
PagerDuty Inc. (PD - $31.57 - Outperform)
Phreesia, Inc. (PHR - $27.38 - Outperform)
Q2 Holdings, Inc. (QTWO - $54.69 - Outperform)
RingCentral, Inc. (RNG - $92.43 - Outperform)
SPS Commerce, Inc. (SPSC - $123.56 - Outperform)
Salesforce.com, Inc. (CRM - $174.57 - Outperform)
ServiceNow, Inc. (NOW - $472.27 - Neutral)
Sprout Social, Inc. (SPT - $65.31 - Outperform)
Veeva Systems Inc. (VEEV - $184.10 - Outperform)
Workiva, Inc. (WK - $105.87 - Outperform)
Zoom Video Communications, Inc. (ZM - $102.44 - Outperform)
Zscaler, Inc. (ZS - $216.79 - Outperform)
nCino, Inc. (NCNO - $38.81 - Neutral)
(See recent research reports for more information)

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